Project Title: AI Autocorrect Tool Using TextBlob

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Problem Statement

Autocorrect systems are widely used in modern applications to detect and correct spelling mistakes in user-entered text. Manual correction can be time-consuming and error-prone. This project develops an AI-based autocorrect tool that automatically identifies spelling errors and suggests corrected text using Natural Language Processing techniques.

Objective

The main objective of this project is to develop an intelligent text correction system capable of detecting and correcting spelling mistakes in user input using the TextBlob library.

Technologies Used

- Python
- TextBlob
- NLTK
- VS Code

3. Dataset Details

Since this project does not require a dataset:

Dataset Name: Not Applicable

Dataset Source: User Input Text

Number of Records: Dynamic User Input

Dataset Link: Not Applicable

4. Project Workflow

1. User enters text.
2. Text is received by the application.
3. TextBlob processes the input.
4. Spelling errors are identified.
5. Corrected text is generated.
6. Original and corrected text are displayed.
7. Correction history is saved.
8. User can continue entering text.
9. Results are stored for future reference.
10. Application exits when requested.

5. Model / Algorithm Used

TextBlob NLP Library

The project uses the TextBlob library for spelling correction.

Why TextBlob?

- Easy implementation
- Built-in spelling correction
- Suitable for beginners
- No large dataset required
- Efficient for text correction tasks

Implementation

Screenshot 1

	Original Text	Corrected Text
0	I hav a dreem to becme an enginer	I had a dream to become an engine
1	Machne learnng is awsome	Machine learning is some
2	Pyhton is my favrite languge	Pyhton is my favorite language
3	I am goin to scholl tomorow	I am going to school tomorrow
4	She dont like eatng vegetabels	The dont like eating vegetables
5	The wether is very pleasent today	The whether is very pleasant today
6	Artifical inteligence is changng the world	Artificial intelligence is changing the world
7	I realy enjoy writting code	I really enjoy writing code

Screenshot 2

```
#Feature 2 : It can count the mistake based on the user input
from textblob import TextBlob

text = input("Enter text: ")

corrected = str(TextBlob(text).correct())

original_words = text.split()
corrected_words = corrected.split()

mistakes = 0

for o, c in zip(original_words, corrected_words):
    if o != c:
        mistakes += 1

print("\nOriginal :", text)
print("Corrected:", corrected)
print("Words corrected:", mistakes)
```

```
Enter text:  my frend helped me a lot

Original : my frend helped me a lot
Corrected: my friend helped me a lot
Words corrected: 1
```

```
Enter text (type 'exit' to quit):  she is veri inteligence person

Original : she is veri inteligence person
Corrected: she is very intelligence person

Enter text (type 'exit' to quit):  That project is veri usefull

Original : That project is veri usefull
Corrected: That project is very useful

Enter text (type 'exit' to quit):  exit
```

Example:

She is veri intelligence person

Corrected output

She is very intelligence person

Important Code Snippet

Use the correct() function:

```
from textblob import TextBlob

text = "I hav a dreem"
corrected = TextBlob(text).correct()
print(corrected)
```

Results

Sample Output

Original Text	Corrected Text
I hav a dreem	I have a dream
Machne learnng is awsome	Machine learning is awesome
Pyhton is my favrite languge	Python is my favorite language

Performance Analysis

The tool successfully corrected common spelling mistakes. However, contextual errors may not always be corrected accurately.

Challenges Faced

- Understanding NLP concepts.
- Installing TextBlob dependencies.
- Handling correction limitations.
- Testing multiple sentence structures.
- Improving correction accuracy.

Conclusion

The AI Autocorrect Tool was successfully developed using Python and TextBlob. The application can identify and correct spelling mistakes in user-entered text. This project demonstrates the practical use of Natural Language Processing in text correction systems.

Future Improvements

- BERT-based contextual correction.
- Grammar checking.
- GUI interface.
- Multi-language support.

References

- https://textblob.readthedocs.io
- https://www.python.org
- https://www.nltk.org

Project

github link : https://github.com/Shreya133-byte/AutocorrectProject_Using_ML